Gamified and Adaptive Learning System Codey for Learning Programming

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***Abstract*— The concept of gamification could be described as the use of game design elements in a non-game context, with the aim that users of a gamified application do not realize that they are doing hard or difficult work but have fun as if they were playing a game. Gamification is one of the most successful approaches in education today, as the use of gamification elements increases students' motivation, engagement and active participation in the learning process, leading to more successful achievement of learning outcomes. This paper presents ongoing research on the use of gamification in education aimed at motivating students and increasing their engagement to improve learning outcomes. The overall aim was to design and develop a gamified system for learning programming and to test its success. The paper describes the self-developed gamified system Codey for Python programming, which includes numerous gamification features to motivate students to master difficult topics related to programming. The most important gamification features implemented are XP (experience points), daily goals, leaderboard, number of days of use (streak), short lessons and code puzzles. Adaptivity was also implemented, i.e. feedback to students after exercises/lessons, repetition of incorrectly answered exercises, adjustment of students' scores, adjustment of the difficulty of tasks and adaptive lessons. The functionality of the system was tested in two phases: Alpha testing to align the intuitiveness of the application with User Experience (UX)/ User Interface (UI) principles, and beta testing with students to test the effectiveness of gamification for learning programming. The research questions were formulated to find out whether the developed gamified system is intuitive to use from a UX/UI perspective and whether the use of a gamified system contributes to greater student engagement in learning programming. The study was conducted in a primary school to test the effectiveness of gamification on student engagement. The results obtained show some evidence of the effectiveness of the use of gamification, but there is a need to continue systematic research on gamification in schools so that the prototype of the gamified system presented is further developed and tested.**

Keywords—gamification, education, adaptive learning, programming

I. Introduction

Gamification is the use of game design elements in a non-game context [1]. It can be used in various fields and the definitions of gamification may differ somewhat, but they all have in common that the reason for using gamification is to motivate user behavior [2]. The same is true for gamification in education, where the goal is to motivate students and increase their engagement [3], [4], but also to improve learning outcomes [5]. Since the feedback students receive during the learning process is one of the most important segments of motivational strategies, the advantage of gamification is that it can provide this in a simple and interesting way for students by turning tasks into games that motivate students through rewards for success [1].

The results of recent systematic literature reviews show that the benefits of using gamification for students are engagement, motivation and creativity. One review [5] analyzed more than 20 research studies comparing the learning outcomes of an experimental group (using gamification) and a control group. It was found that academic results were significantly better in the experimental groups. However, both earlier research [2] and more recent research [3], [6], [7] confirm that the levels of education that frequently include gamification in their curricula are higher education institutions and much less so primary and secondary schools. It also shows that a comprehensive overview of empirical evidence for the effectiveness of gamification in K-12 education is still required.

This paper contributes to research in the field of gamification in education by developing its own gamified system for learning programming in Python [8]. The aim was to develop a system that would maintain students' interest and attention while learning a subject that proves to be difficult [9]. The gamified system called Codey contains many gamification elements such as points, leaderboards, challenges, a quiz system, puzzles and adaptation. In the continuation of the research, the gamification elements of the system were analyzed and improved based on the originally proposed gamification framework [11]. The system was tested to eliminate errors and increase the intuitiveness of the user interface. In addition, the system was tested in a real school environment to evaluate the effectiveness of gamification. The results obtained demonstrate the potential effectiveness of using gamification to increase student engagement and will be used for further research related to the application of gamification in education.

II. Related Work

To identify the desirable characteristics of a proprietary gamified system, existing digital tools for gamification were analyzed. A systematic literature review [6] found that schools use both their own gamification applications and publicly available tools. Kahoot!, Moodle, Edmodo, PeerWise, Los Cokitos, and ClassDojo are among the tools that enable the

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creation of gamified teaching content in various subjects. Some of the tools are available for teaching content in specific areas, e.g. Reading Battle for gamification in foreign language learning or Ocean Literacy for the field of biology. The proprietary gamified mobile learning platform Math Widget is described in [10] and is intended for practicing math content in lower primary school classes. The authors conducted research with various gamified learning activities, including adaptive ones, and found that the performance level of students who used adaptive gamification improved.

In the research described in [11], 9 publicly available digital tools for gamification were selected and analyzed using the proposed classification based on the MDA framework [12], [13]. 23 elements of gamification were selected as the most important for education and categorized into 4 categories:

- *Mechanics*: Points, Badges, Leaderboards, Levels, Progress bar;
- *Dynamics*: Prizes/rewards, Feedback, Adaptation, Puzzle, Quiz system, Synchronous use, Asynchronous use;
- *Aesthetics*: Avatar, Live reactions, Challenge, Competition, Cooperation, Storyline/narrative, Fantasy;
- *Other*: Reports, Lessons, Classes, AI tools.

During the preliminary research, systems and tools for learning programming were also examined [8], some of which are more concerned with the use of game-based learning and/or adaptability than with gamification, but they are nevertheless useful from the point of view of applying game design elements.

The Hour of Code is a free introduction to computer science through fun activities and videos for students of all levels. The learning process takes place in a playful way. The student can choose a topic of their choice, and the topics are usually related to some famous games or films [14].

Codecademy is an interactive platform for learning programming and markup languages [15]. Apart from basic game design elements such as points and badges, Codecademy does not have many gamification elements, but it does offer some adaptability to users.

Lightbot is an educational video game designed to teach programming concepts "secretly" by the player "while playing the game [16]. Although this application is an educational game and is more complex than gamified systems, it is a good example of how basic programming concepts can be learnt in a simple way.

The analyzed systems and tools for gamified learning of programming as well as those that implement gamification, proved to be valuable examples that enable the design and implementation of the self-developed gamified system Codey.

III. Methodology

The overall objective of the research was to design and develop a gamified system for learning programming and to test its success. The functionality of the system was tested in two phases: alpha testing to align the intuitiveness of the application with User Experience (UX) / User Interface (UI) principles, and beta testing with students to test the effectiveness of gamification for learning programming. The objective was achieved by answering the following research questions:

RQ1: Is the developed gamified system intuitive to use from a UX/UI perspective?

RQ2: Are students who use gamified system more engaged in learning programming?

A. Gamified System Codey

The developed gamified system Codey consists of a client web application built using a Flutter framework, and a corresponding server (.NET), with a MongoDB data store. The application was no longer publicly accessible at the time this paper was written.

The application was designed to support four different types of users: students, teachers, creators and administrators. Students using the application can be assigned to classes by a teacher or use the application without an assigned class. The programming course was designed with three levels of granularity – units, lessons, and exercises. The prototype course included in the system currently consists of four units: *We print things*, *Variables*, *Data input*, and *Calculation*. Each unit contains several lessons (usually 3-4), while each lesson contains several exercises (usually up to 10).

Four exercise types were implemented: multiple-choice, short answer, writing parts of code, and writing a complete program (Fig. 1). To make it easier to write a complete program, a code puzzle was implemented that offers both correct and incorrect pieces of code that can be tapped to assemble the answer.

On the initial screen, students can see their progress in the course, through the perspective of units. The course must be solved sequentially, but previously completed units can be revisited. The lessons in the units follow a similar sequential approach. Students can start solving the exercises immediately, or they can read tips on the content of a unit or lesson. The exercises in a lesson are arranged in a way to enable students to learn the intended concepts gradually, where the first few exercises are very easy, and then the exercises get progressively more difficult.

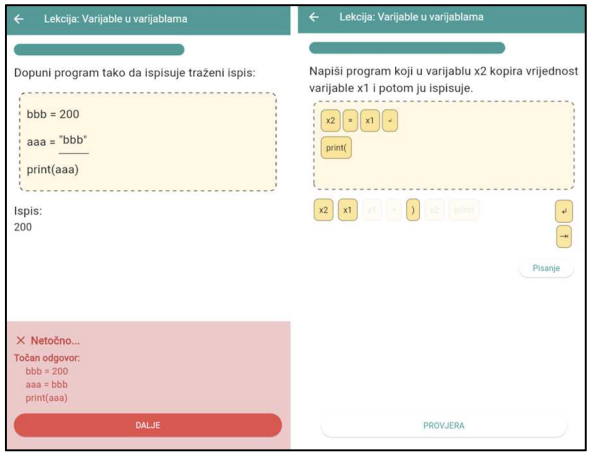

Fig. 1. Short and long code writing exercises

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If the student answers an exercise correctly, they can continue with the next exercise until the end of the lesson is reached. If the answer is incorrect, the expected correct answer is displayed. To achieve spaced repetition, all incorrectly answered exercises are repeated at the end of the lesson. The exercises are repeated until a correct answer is entered.

1) Gamification Features of the System

Gamification features implemented in the Codey are summarized in the Table 1.

Students earn experience points (XP points) by completing lessons. They receive more XP points for completing a new lesson (100XP) and fewer for repeating a lesson they have already completed (40XP).

Each day, students are given two goals or daily quests to complete as a challenge. For each completed goal they are awarded 50 XP. Examples of implemented tasks are: high accuracy (complete 2 lessons, each with an accuracy of > 90%), high speed (complete 2 lessons, each in less than 45 seconds), complete a unit and earn a certain number of XP points.

After completing each lesson, students can see a leaderboard on their profile screen ranking them by total XP points earned. This feature allows students in the same class to compare and compete with each other. Each day a student completes a lesson, their streak (consecutive days of completion) is extended. This feature can encourage students to study daily.

When solving the tasks, students can not only write the answers manually, but also put them together from smaller pieces that resemble a game (puzzle).

On the achievement screen, students can see their total XP points, their streak, the daily challenges and the leaderboard of all students in their class (Fig. 2).

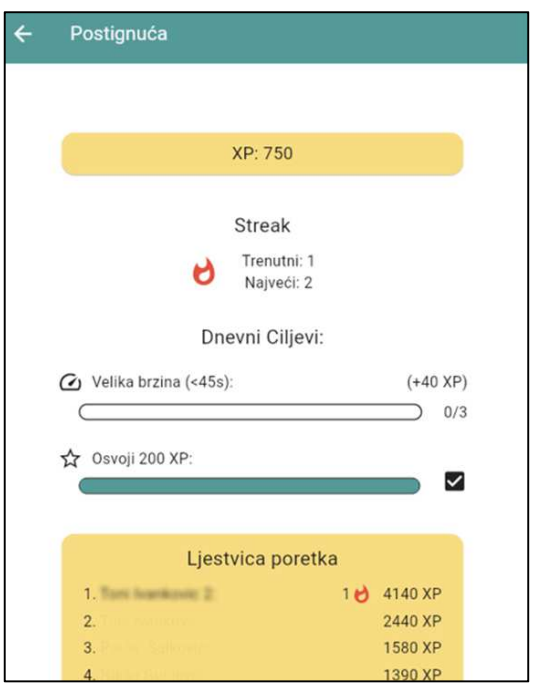

Fig. 2. The achievement screen

TABLE I. IMPLEMENTED GAMIFICATION FEATURES

Feature	General description	Description in the application
Mechanics		
Points	Users can collect numerical points that are awarded for achievements.	Experience points (XP)
Leaderboards	Ranking of users based on the results achieved, mostly points accumulated.	Class leaderboard (or the entire school leaderboard, if no class is assigned)
Levels	Completing a certain number of tasks, users are allowed to move up to the next level.	Unit/lesson progression
Progress bar	It shows the user's progress and guides them through the system.	Progress within a lesson
Dynamics		
Feedback	Quick feedback on the user's current engagement.	Feedback on every exercise answer, feedback at the end of the lesson
Adaptation	The system changes depending on the user's actions	Adaptive lessons
Puzzle	Solving additional games, puzzles, etc.	Code writing puzzle (selecting correct code pieces)
Quiz system	Solving quizzes with different types of questions, usually multiple choice.	Exercises of different types
Asynchronous use	Gamification during the user's independent learning.	Independent learning in class or at home
Aesthetics		
Challenge	An element for encouraging the user to perform tasks.	Daily quests, streak
Competition	Competition with other users using gamification elements.	Performing on the leaderboard
Other		
Reports	The teacher has access to student data on participation in gamified activities.	Viewing the class leaderboard and individual student data
Lessons	Lessons can be created as presentations and training materials.	Units/lessons as grouping categories, unit/lesson hints
Classes	The teacher can create virtual classes and enroll their students.	Grouping students in classes

Special attention was paid to the gamification element adaptation from the Dynamics category. Certain lessons are adaptive, and 15 exercises are selected from the set of all solved exercises that the student has learnt, based on the student's internal score. The internal score is adjusted each time the student answers an exercise.

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If the student provides a correct answer to an exercise with a difficulty larger than the student's current score, the student's score is increased. Similarly, if the student provides an incorrect answer to an exercise with a difficulty smaller than the student's current score, the student's score is decreased. When selecting exercises for an adaptive lesson, exercises with a level of difficulty corresponding to the student's score are selected. In addition, of the 15 exercises selected, 5 were chosen with a lower difficulty level than the student's current score (to test existing knowledge) and 10 with a higher difficulty level than the student's score (to provide the opportunity to learn and practice more difficult concepts).

B. *Participants and Study Design*

To answer the RQ1 "Is the developed gamified system intuitive to use from a UX/UI perspective?", the system was tested with six university students aged 22-25, after all intended features were implemented. Three testers were selected from the computer science study, with extensive Python programming knowledge, and three testers were selected from unrelated STEM study fields, with no prior programming knowledge. Testers with a computer science background were selected to provide insights from a technical perspective, while other testers focused more on the course content and had a perspective more similar to the intended users of the application.

The users were instructed to follow a questionnaire while using the application, rating the intuitiveness of the interfaces in the application, and the visual appeal. At every step, the testers were asked to suggest possible alterations to the application. They were also asked to approximate how much gamification features contributed to their motivation and how likely they would hypothetically use the application for the purpose of learning programming.

To answer the RQ2 "Are students who use gamified system more engaged in learning programming?", a beta testing was conducted at primary school with fifth-grade students who had no previous knowledge of Python programming. The test was conducted with 48 students, 25 of whom were placed into a gamified group and 23 into a control group that received the application without the insight into XP, daily quests, leaderboard and streak. Allocation to the groups was randomized (determined by the order of registering an account), and the uneven number of students occurred because some students did not participate in the study. The students were asked to use the system in their computer science class.

In order to examine whether gamification increases student engagement, the total number of lessons solved, the number of new lessons solved (course reach), the time spent in the lessons, the level achieved (student score), and the accuracy of solving the lessons were analyzed.

IV. Results and Discussion

A. *Application Intuitiveness Testing*

The intuitiveness of the introduction to learning with the application was rated by all testers with 5 out of 5 points. However, they did point out that the button for starting the lesson should be formatted more clearly and that the lesson hints button should be described with text (it was originally formatted as an icon).

The intuitiveness of the learning interface (within the lessons) was rated 5 out of 5 points by 83% of the testers. The level of difficulty of the first lesson of the course was rated as sufficient or slightly too easy by all testers. The interactivity was rated 4 out of 5 points by 67% of the testers.

When asked to approximate how likely they would be to use the app for learning, the results varied, with a third of respondents approximating the likelihood at 2-3, a third at 5, and a third at 7-8 out of 10.

However, when asked to approximate how gamification features contributed to their motivation, more than 80% of the respondents placed the contribution at 7-9 out of 10.

Generally, the respondents rated the intuitiveness of the application UX/UI positively, but some improvements were suggested, which were mostly implemented before the beta testing part. Namely, more hints were added on specific exercises, more trick-questions were added to make the content more challenging, the student's name on the leaderboard was bolded to stand out between more users, and the UI design was unified throughout the application.

Besides the fact that the alpha testing helped to further improve the application from the UX/UI aspect, the main question of interest for testing the system in the next phase was related to the subjective evaluation of the gamification elements (XP points, leaderboard, daily challenges, streak, code puzzle, etc.). Therefore, it can be considered satisfactory that the majority of testers believed that the gamified application could have a significant impact on increasing motivation to learn programming.

B. *Learning engagement*

When the beta testing with students was finished, application logs were used to analyze 5 variables for each student – the total number of lessons solved, course reach, internal student score, the time spent in lessons, and the accuracy of solving the lessons.

Considering the sample size, the assumption of normal distribution of the data was not fulfilled, so that the non-parametric Mann-Whitney U test was used. Therefore, the medians of the variables are presented, but the distributions of the data also play a role in determining the p-value. The data obtained are presented together with the statistical test data in Table 2.

TABLE II. Results of logs analysis

Variable	Cont. group	Gam. group	p-value
Total lessons solved	6	7	0.064
Course reach	6	6	0.134
Student score	1.32	1.45	0.179
Median time spent in lessons	276.27	255.31	0.241
Median solving accuracy	82.21 %	76.92 %	0.163

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The results show that the total number of lessons solved is slightly higher in the gamified group than in the control group with a relatively small p-value. The students using gamification scored higher than the control group (but with a p-value of 0.179). The students using gamification solved the lessons faster, but this conclusion has the lowest statistical significance with a p-value of 0.241. Surprisingly, the students with gamification had a lower solution accuracy.

The results obtained only partially support the claim associated with Q2 that the use of a gamified application contributes to higher student engagement. The lack of normality of the data (and the use of non-parametric statistical tests) as well as the limited sample size resulted in relatively high p-values, with the exception of the total number of lessons solved, which was measured higher in the gamified group with a p-value of 0.064. The results may have been largely influenced by the design of the experiment – within a single class period, several gamification features offer very limited differences in experience (e.g., streak, daily quests). In addition, research has not yet investigated whether increased engagement can further strengthen students' motivation to learn programming.

V. Conclusion and Future Plans

This paper describes a gamified system for learning Python programming called Codey. The first part of the course was created as a prototype designed for learning data printing, working with variables, data entry and arithmetic operations. It contains many gamification elements that have been added to increase student motivation and improve the learning experience.

The application was tested by 6 testers to identify possible improvements in terms of design and UX/UI intuitiveness. In addition, the research was conducted in school on a sample of 48 students and showed results that signal the effectiveness of gamification for the purpose of student motivation. However, the results are inconclusive, possibly due to the insufficiently large sample size and the experimental setting in which gamification could not achieve its full effect.

These results provide a good basis for further research into the use of gamification in schools. It is necessary to further develop the prototype of the system to include more lessons and to repeat the study on a larger group of students and over a longer period of time. In addition to analyzing the data from the application logs, it is necessary to conduct a survey among the students to determine whether their motivation to learn programming has increased.

In the continuation of the research within the project "Enhancing Motivation for Learning Through Gamification", it is planned to develop new technological solutions for gamification that can be used in other subjects in primary and secondary school. The effectiveness of different combinations of game elements will be analyzed in order to develop a specific pedagogical-technological framework for gamification and recommendations for teachers to help them design and implement gamified activities for students.

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